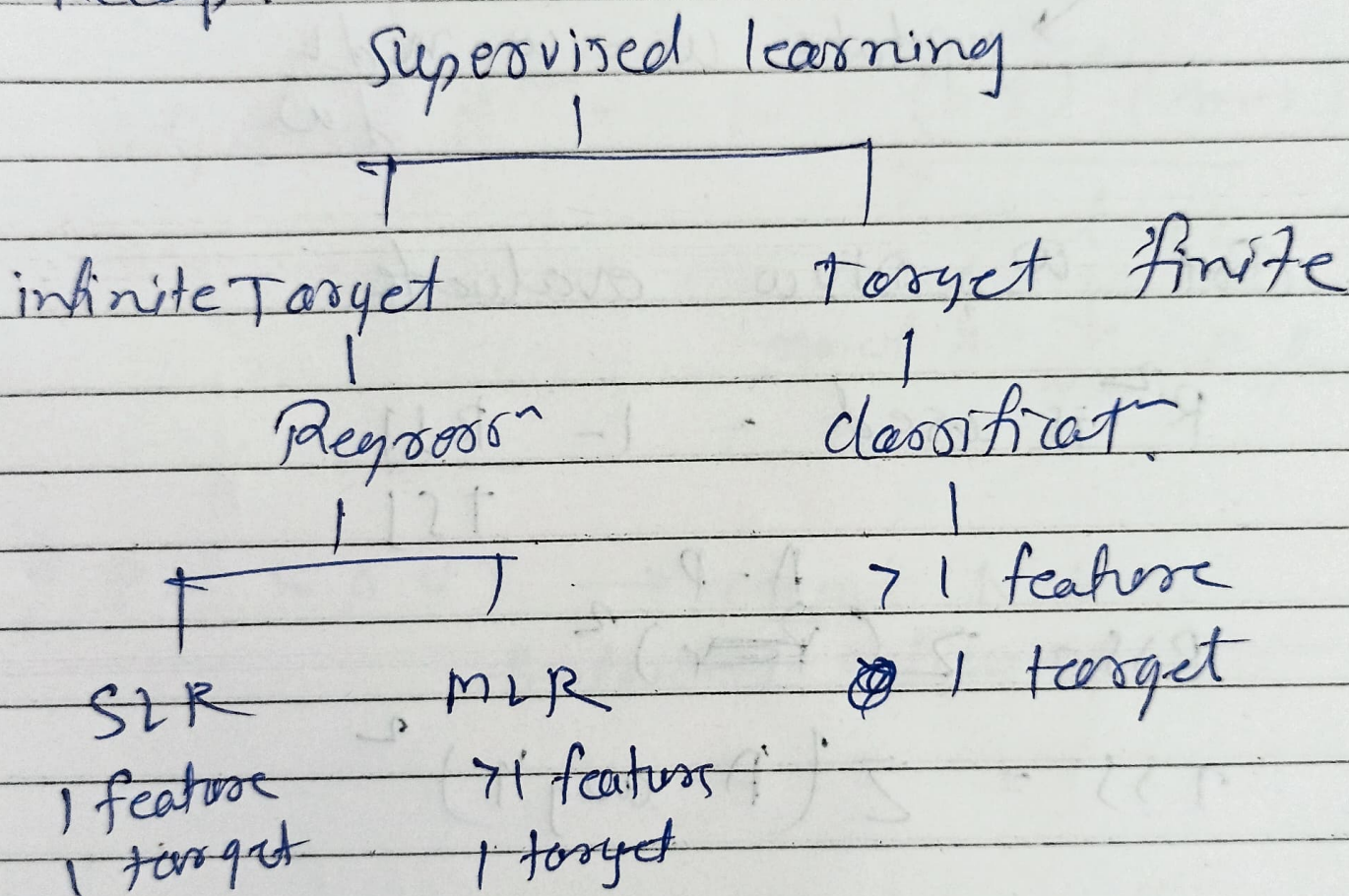


25-ML-LR4 :- Assumptions

Recap :-



Linear -> stepcion -> fit -> do not want more stats

OLS -> fit -> want more stats.

form scratch

~~used GD -> need Loss function~~

GD $\rightarrow$ Loss $\rightarrow$ MSE

$\rightarrow$ update $w = w - \eta \frac{dL}{dw}$

Got w , Now evaluate

R^2 is used $= 1 - \frac{RSS}{TSS}$

$RSS = \sum (A - \hat{P})^2$

$TSS = \sum (A - \text{avg } Y)^2$

$R^2 = \{0, 1\}$ practical

$= \{-\infty, 1\}$ theoretically

R^2 can not go down by increasing feature

$$\text{Adjusted } R^2 = 1 - \frac{(1-R)^2 (m-1)}{(m-d-1)}$$

no. of X no. of feature

$p < 0.05 \rightarrow$ reject Null
feature is relevant

Assumption

① No multi-collinearity

age	year	make	model	...	price
12	2012				
17	2010				
16	2008				
27	2000				

age ↑ years ↓

age & years are co-related

x	y	$z = x + y$	...	price

$$z = w_1 x + w_2 y$$

$$w_1 = w_2 = 1$$

$$z = x + y$$

this is called multi-collinearity
multi-collinearity is big problem
of LR.

$$y = m_1 x_1 + m_2 x_2 + m_3 x_3 + m_4 x_4 + z$$

$\downarrow$
 depedent are independent
 variable

~~i.e. no multi-collinearity~~

if $x_1 = x_2 + x_3$

x_1 is not independent

I think this is called multicollinearity

* ~~Why~~ Why multi-collinearity is an issue

$$y = w_1 x_1 + w_2 x_2 + w_3 x_3$$

$$w = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

so here x_3 is most imp. feature

$$\text{Let } x_2 = 1.5 x_1$$

$$w_1 x_1 + w_2 x_2 + w_3 x_3$$

$$1x_1 + 2x_2 + 3x_3$$

$$x_1 + 2 \times 1.5 x_1 + 3x_3$$

$$4x_1 + 3x_3$$

$$w = \begin{bmatrix} 4 & 0 & 3 \end{bmatrix}$$

x_1 is more imp. feature.

Multi-collinearity will not be problem to fit nework, It will give w but messes up. with feature ~~interpretation~~ interpretation.

* How to identify multi-collinearity in data

* How to solve it

Multi-collinearity considers only independent X features, No target y

* How to find multi-collinearity

suppose

$$\begin{array}{c|c} f_1 & f_2 & f_3 & f_4 & y \end{array}$$

Here I want to find something

$$f_1 = \omega_1 f_2 + \omega_2 f_3 + \omega_3 f_4$$

$$\rightarrow f_2 = \omega_1 f_1 + \omega_2 f_3 + \omega_3 f_4$$

$$f_3 = \omega_1 f_1 + \omega_2 f_2 + \omega_3 f_4$$

$$f_4 = \omega_1 f_1 + \omega_2 f_2 + \omega_3 f_3$$

Do LR ~~on~~ on f_1

this will give you R^2 which tell how the model is.

$$\text{If } R^2 = 0.8 = 1 - \frac{RSS}{TSS}$$

Then 80% f_1 can be got from f_2, f_3, f_4

Calculate R^2 for all f_1, f_2, f_3, f_4

$$VIF = \frac{1}{1 - R^2}$$

$$VIF_{f_1} = \frac{1}{1 - 0.8} = 5$$

take some threshold for VIF say 5
if $VIF \geq 5$ then reject f_1

Threshold is different for different scenarios.

If VIF of different features

$$f_1 = 7.9$$

$$f_2 = 8.3$$

$$f_3 = 0.44$$

$$f_4 = 0.9$$

$$f_5 = 0.7$$

then whether f_1 & f_2 should be removed together or only f_1 or f_2

So removing of features by looking VIF should happen one by one.

case I

$$R^2 = 1$$

$$VIF = \frac{1}{1 - R^2} = \infty$$

highly correlated

Remove

case II

$$R^2 = 0$$

$$VIF = \frac{1}{1 - R^2} = 1$$

not correlated

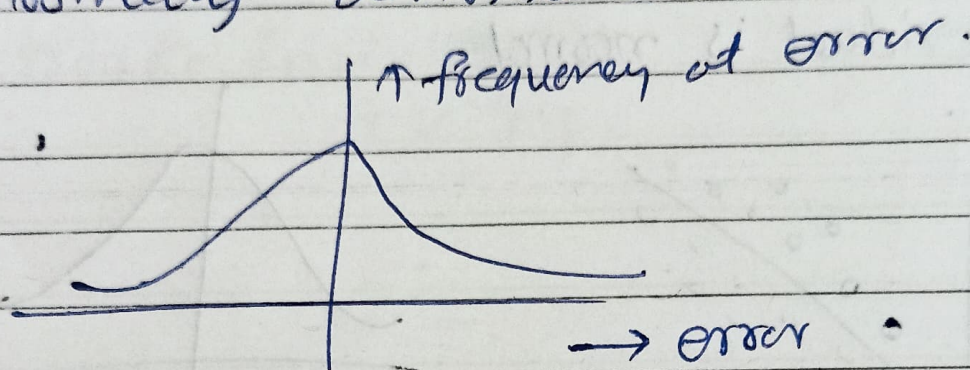
do not remove

look at "15-m2-LR4-VIF-implementation"
"ipynb"

~~for~~

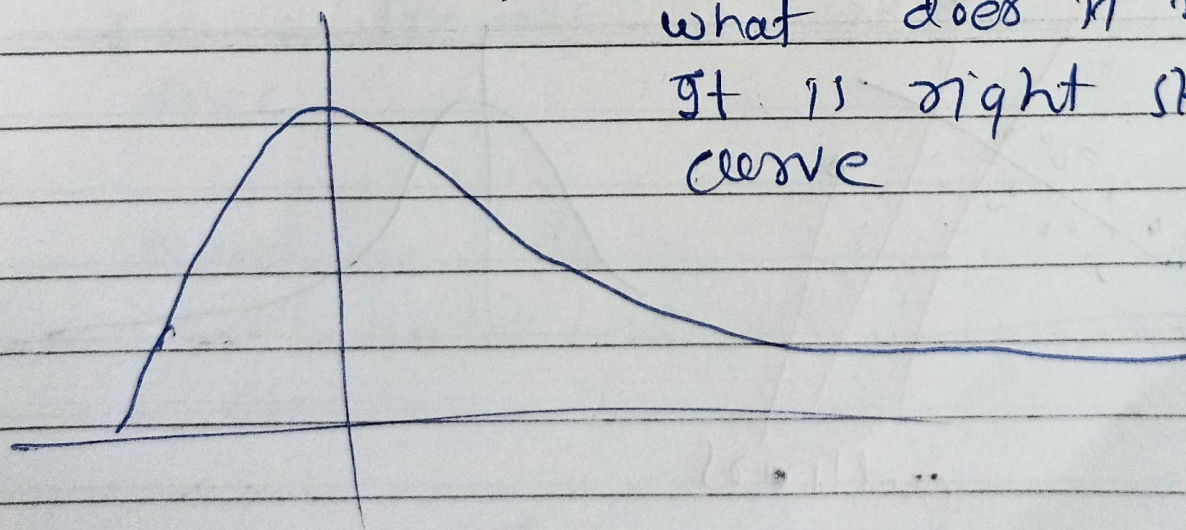
Assumption

- ② The residual / error should be normally distributed



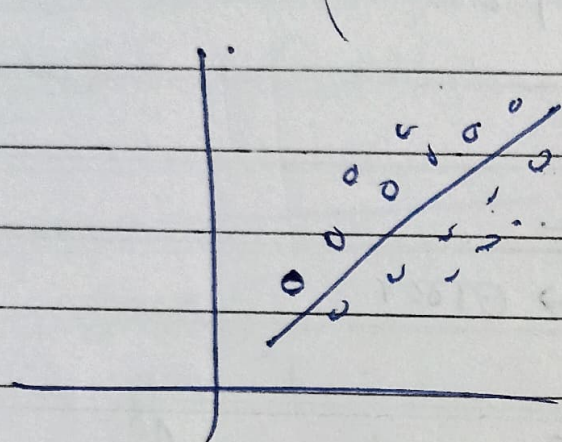
0 error will be more, frequency of should decrease when error ↑.

If the distribut of ~~residual~~ residual is something like below what does it mean
It is right skewed curve

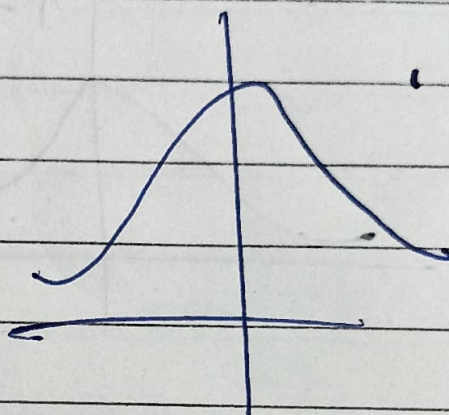


This happens because of outliers
in data

when residual is normal

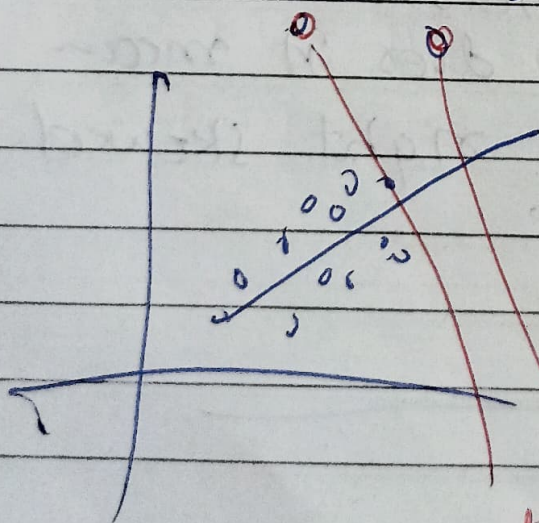


graph

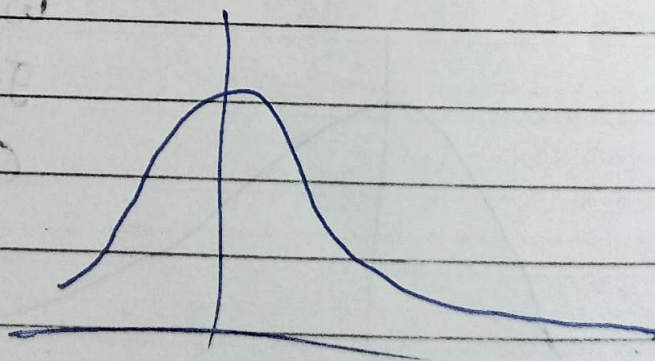


Residual

when residual is right skewed

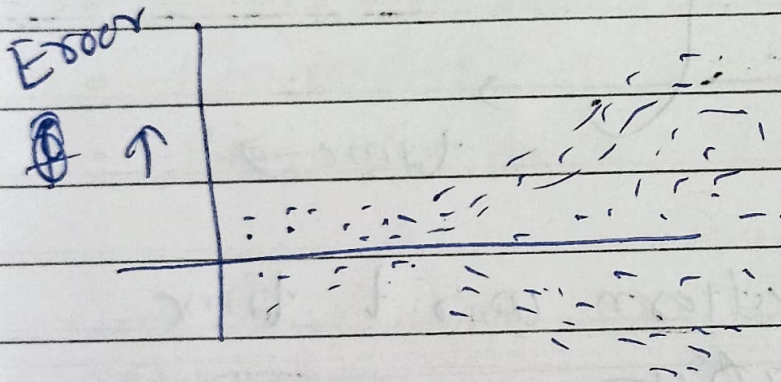


outliers



Same is true for left skewed.

③ heteroskedasticity

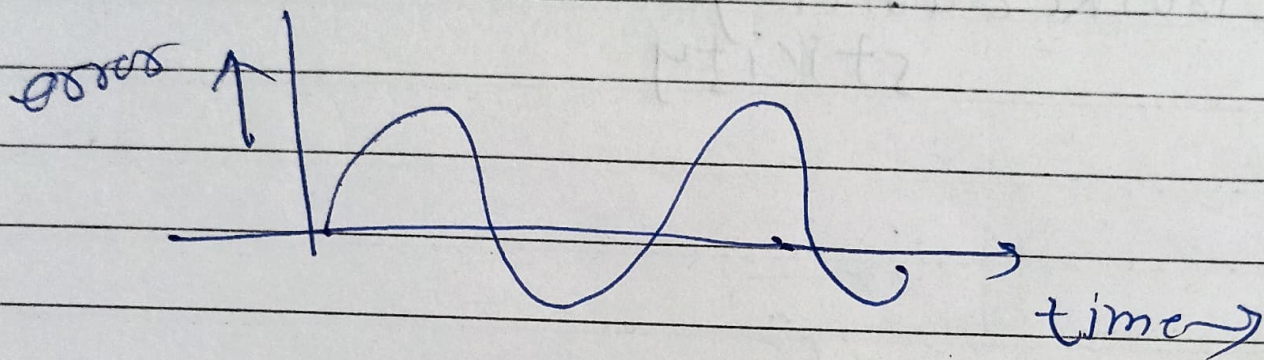


→ ~~error~~ house price (\$)

model is giving different error
for different range of house
price

Need to change model / or its
not linear model, or anything
else.

④ Auto correlation
self correlation



if errors have ^{any} pattern w.r.t. time
then auto correlation

Then LR will not work.

for LR there should not be any
pattern.